

DATABRICKS WORKSHOP

Interactive Architecture Design Workshop

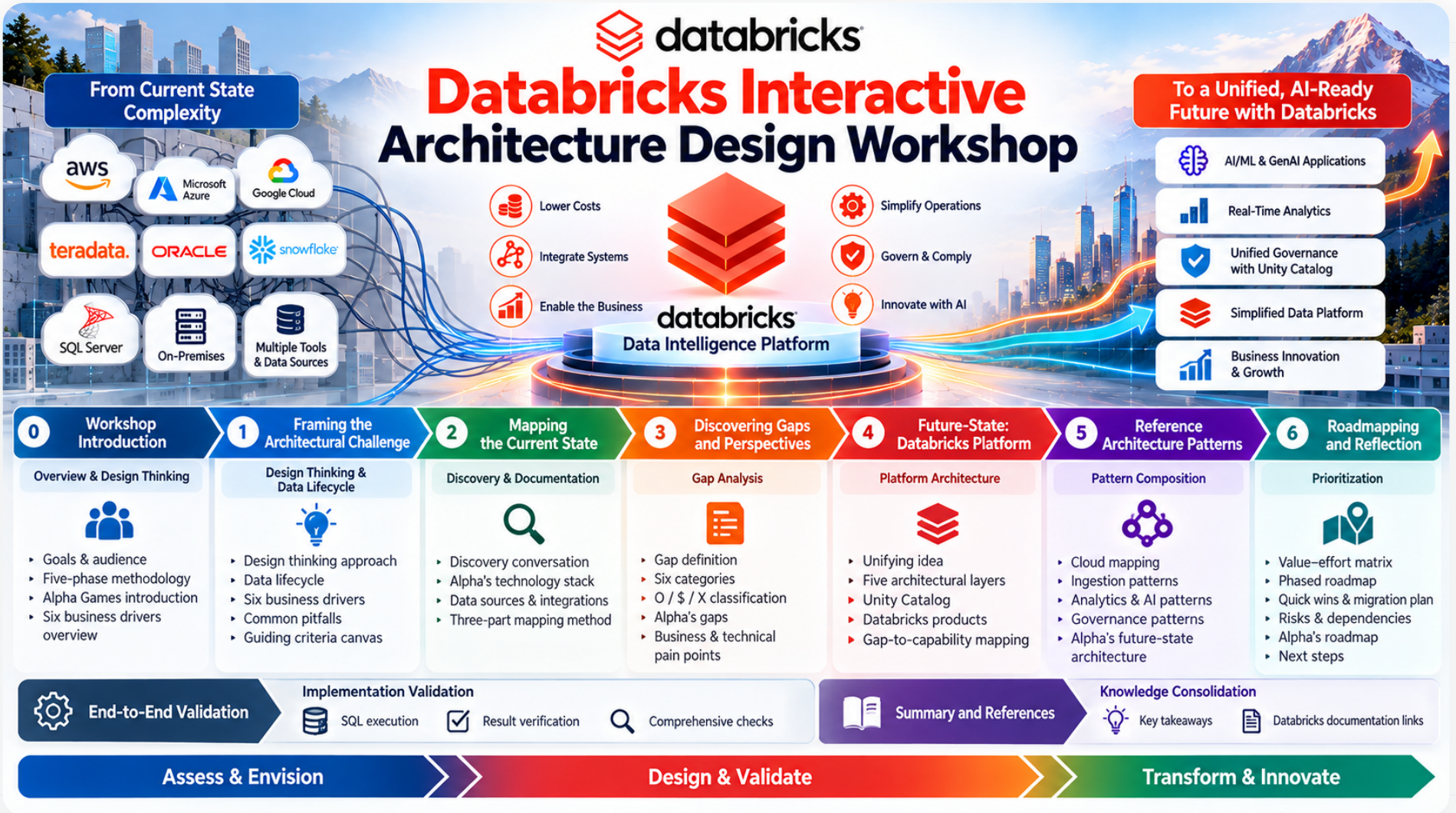
A design-thinking approach to discovering, designing, and roadmapping a modern data architecture on the Databricks Intelligence Platform.

Solution Architects

Data Leaders

Platform Engineers

Yogesh Ghorpade • Databricks



End-to-end workshop roadmap — from discovery to implementation across six structured modules.

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Databricks Interactive Architecture Design Workshop



Module-by-module flow showing how each design-thinking phase produces a concrete deliverable.

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WORKSHOP FLOW

Six Modules. One Outcome.

Each module builds on the last — from understanding the business to delivering a phased roadmap.

MODULE 1

Framing the Challenge

Identify business drivers, rank priorities, and define guiding criteria for architecture decisions.

MODULE 2

Mapping Current-State

Document existing systems, data flows, and technology inventory across the full data lifecycle.

MODULE 3

Discovering Gaps

Classify gaps as Opportunities, Cost Optimizations, or Critical Risks using the O/\$/X framework.

MODULE 4

Databricks Platform

Map gaps to Databricks capabilities. Understand Unity Catalog, Lakeflow, SQL, AI/BI, and AI&ML.

MODULE 5

Reference Patterns

Compose proven ingestion, analytics, and AI patterns into a future-state reference architecture.

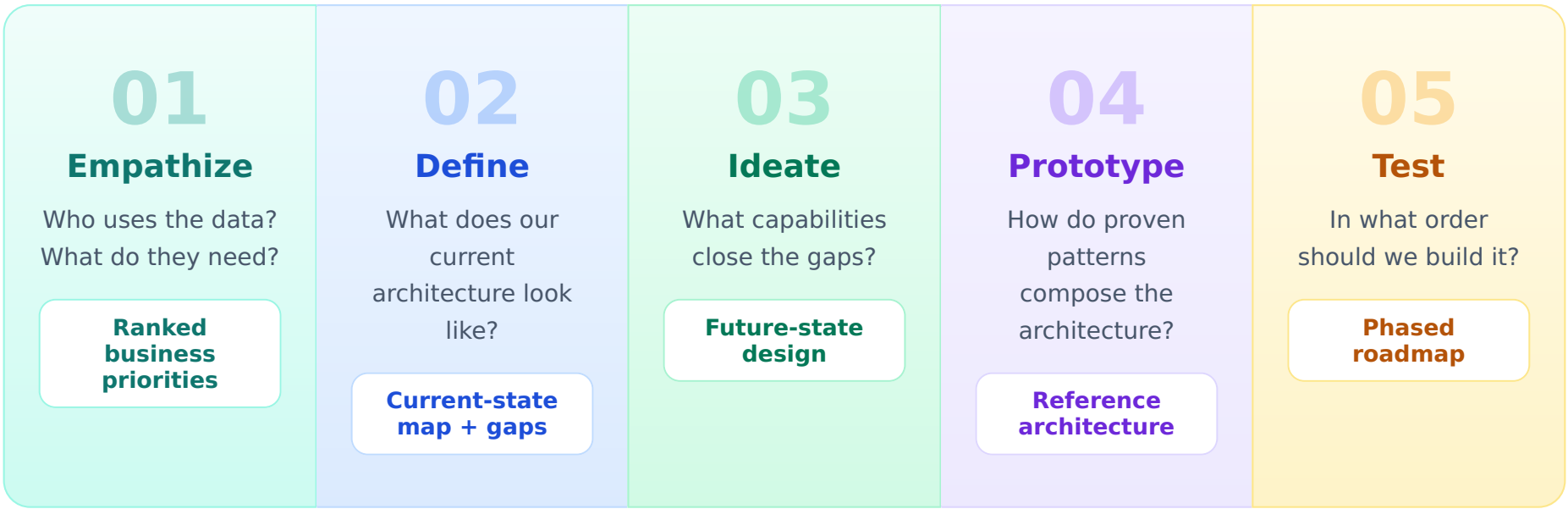
MODULE 6

Roadmapping

Prioritize initiatives on a value-effort matrix and sequence them into a phased delivery roadmap.

Design Thinking for Data Architecture

Five phases. Each builds on the last. Skip one and the whole design weakens.



Outcome flow: Business context → Current-state truth → Platform options → Architecture patterns → Execution plan

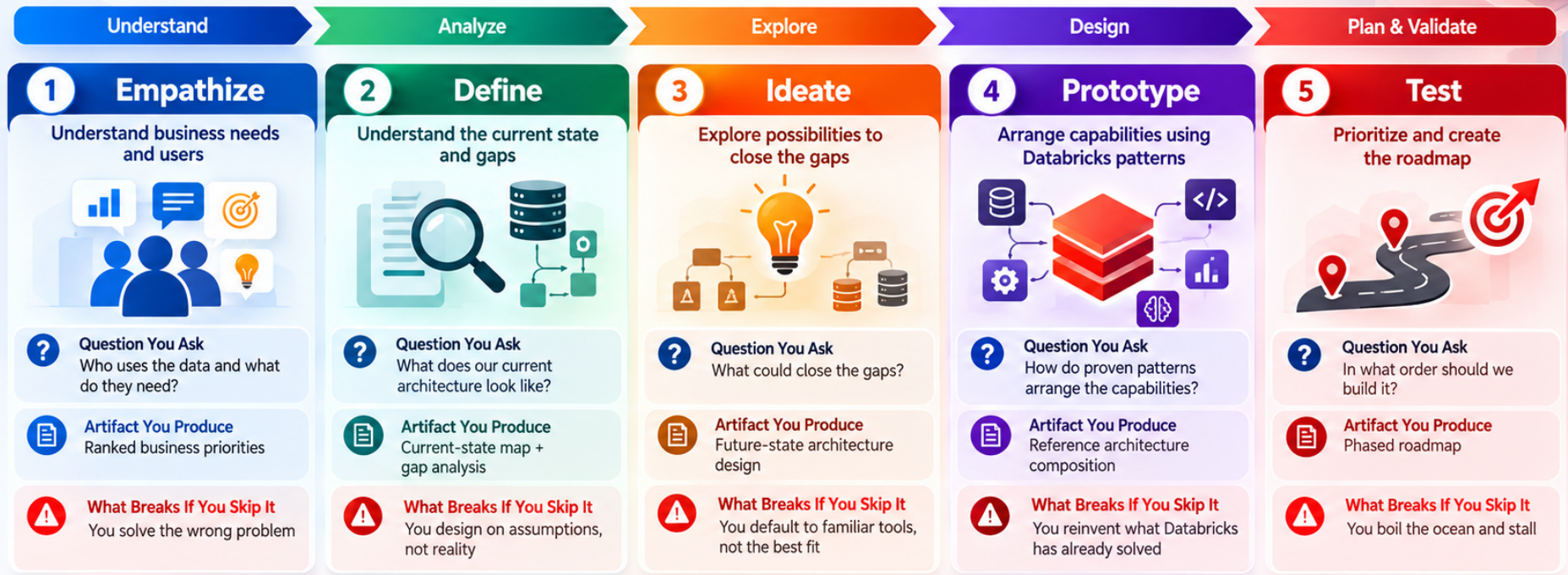


Databricks Interactive Architecture Design Workshop

From
Business Needs
to a Unified,
AI-Ready Future



Design Thinking: The Five Phases



From Design Thinking to a Databricks Architecture



The five design-thinking phases applied to data architecture — from empathy to execution.

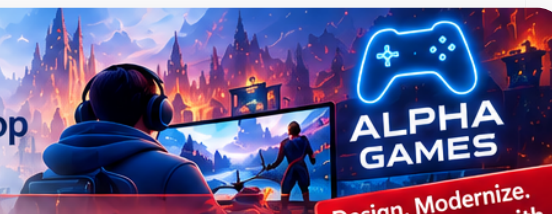
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Workshop Introduction

Databricks Interactive Architecture Design Workshop

From Business Needs to a Unified, AI-Ready Data Architecture



01 What This Workshop Is



The Databricks Interactive Architecture Design Workshop is an instructor-led discovery session that applies **design thinking** to data architecture planning. Participants assess their current environment, identify challenges, and design a future-state architecture on **Databricks** aligned with organizational business needs.

Target Audience



Data architects, data engineers, and technical decision makers



IT leaders responsible for data strategy and platform decisions



Technical business stakeholders involved in data modernization



Individuals evaluating Databricks for their organization



Workshop Goals

1

Apply design thinking principles to assess and reimagine data architecture

2

Document and analyze current-state architecture across the data lifecycle

3

Identify and evaluate gaps and optimization opportunities

4

Design a future-state architecture integrating Databricks components

5

Develop a phased implementation roadmap aligned to business goals

6

Communicate architectural reasoning effectively with stakeholders

Design. Modernize. Unify. Innovate with Databricks.

Workshop Flow: From Current State to Future State

A structured, interactive journey using **Design Thinking** for architecture design

1

Empathize



Understand business needs and users

Who uses the data and what do they need?

2

Define



Assess current-state systems and gaps

What does our current architecture look like?

3

Ideate



Explore possibilities to close the gaps

What could close the gaps?

4

Prototype



Compose proven reference patterns into architecture

How do we arrange the capabilities?

5

Test



Prioritize delivery into a phased roadmap

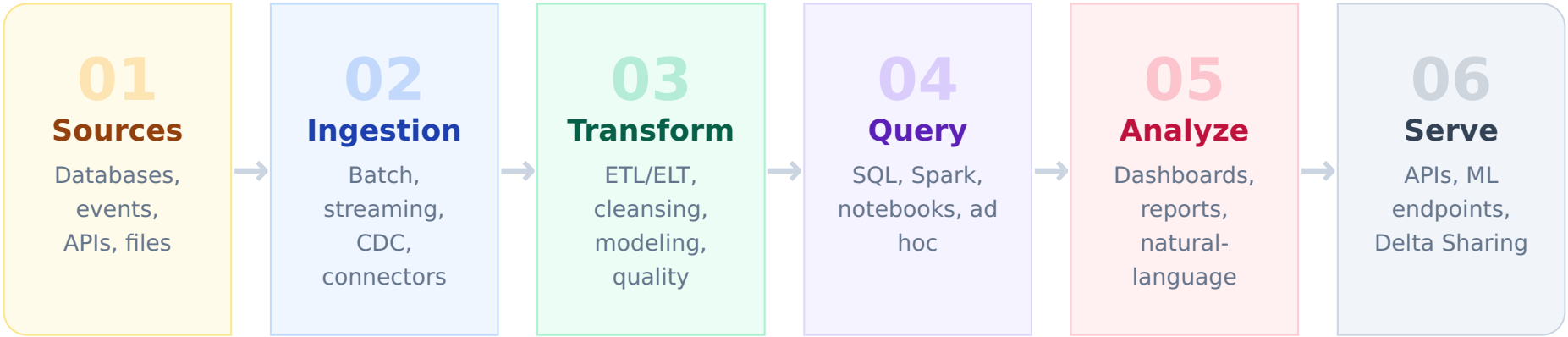
In what order should we build it?

Workshop target audience, goals, and the design-thinking approach to architecture planning.

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The Universal Data Lifecycle

Every data architecture moves through these six stages. A weakness in one becomes a bottleneck in the next.



RUNNING EXAMPLE

Alpha Games

A fictional gaming studio migrating to the cloud. Their data stack grew tool by tool and now strains under scale. We follow Alpha through every design-thinking phase.

- Operational intelligence for analysts
- AI serving for in-game recommendations
- Real-time player behavior analytics (future need)

CURRENT STACK

- S3 • Raw object storage
- Python scripts • DMS • CSV uploads
- EMR • Separate ETL server
- Redshift • Separate data copy
- Tableau • SageMaker (standalone)

Current-State Architecture

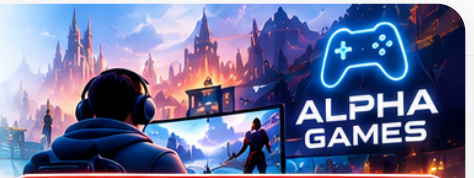
Data flows through disconnected tools with fragile handoffs at every stage.





Use Case: Alpha Games — The Running Example

A fictional gaming studio migrating to the cloud. Their data stack grew tool by tool over the years and now strains under scale. The workshop follows Alpha through every phase of design thinking.



Alpha Games

A fictional gaming studio migrating to the cloud. Their data stack grew tool by tool over the years and now strains under scale.



Current State

- Runs on AWS with two primary use cases: operational intelligence for analysts and AI serving for an in-game recommender
- Uses a patchwork of tools: S3 for raw storage, multiple ETL scripts, a separate warehouse, a BI tool, and a standalone ML platform
- Data flows through manual pipelines with fragile handoffs between tools



Future Needs

- Real-time player behavior analytics for game designers
- A unified governance layer to meet new regulatory requirements
- Ability to deploy and serve ML models without a separate platform

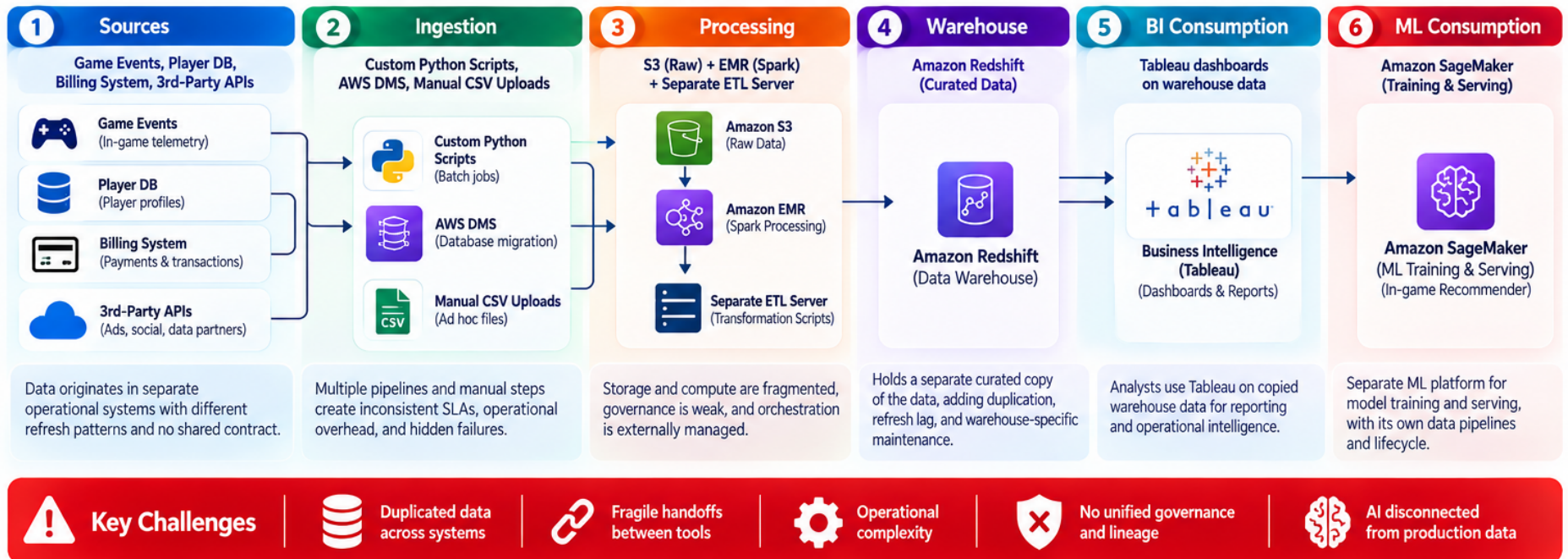


Current State Impact

- Duplicated data across multiple systems
- Fragile orchestration and manual handoffs
- Higher operational overhead
- No unified governance and lineage
- AI disconnected from production data

Alpha Games: Current-State Architecture (AWS)

Current flow shows why Alpha experiences duplication, fragile handoffs, and disconnected ML operations.



Alpha Games current-state AWS architecture — fragmented tools, duplicated data, and no unified governance.

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Architecture Design

Driven by Business Outcomes



Six Business Drivers

1



Reduction of Costs

Lower TCO by consolidating tools.

Alpha Example:

Paying for EMR + Redshift + SageMaker + Tableau separately.

2



Integration

Connect siloed systems.

Alpha Example:

Manual data movement between S3, Redshift, SageMaker.

3



Enabling the Business

Support new use cases.

Alpha Example:

Real-time player analytics needed but pipelines are batch-only.

4



Simplification

Reduce operational complexity.

Alpha Example:

7+ tools requiring separate maintenance and skills.

5



Governance & Compliance

Data security and regulatory compliance.

Alpha Example:

No unified access control or audit trail.

6



Innovation & AI

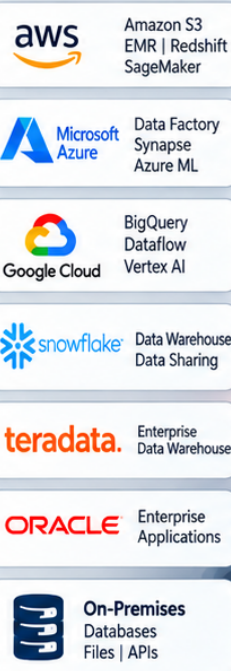
Advanced analytics and AI capabilities.

Alpha Example:

In-game AI needs but ML platform is standalone.

Before: Multiple Platforms

After: Unified with Databricks



databricks
Unified Data Intelligence
Platform



Common Pitfalls



High Tool Sprawl

Managing multiple tools increases cost and complexity.



Siloed Data

Data stuck in different systems with limited integration.



Delayed Insights

Batch-only pipelines limit real-time decision making.



Operational Complexity

Multiple tools, teams and skill sets to maintain.



Weak Governance

No unified access control, lineage or audit trail.



Limited AI Innovation

Standalone ML platforms slow down advanced analytics and AI use cases.

From Disconnected Tools & Complexity

To a Unified, Governed and AI-Ready Databricks Platform

Every architecture component traces back to a business outcome — drivers shape use cases shape components.

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1 What is Design Thinking?



Design thinking is a practical, creative method for solving complex problems by balancing three forces: **empathy** for the people who have the problem, **creativity** to generate solutions, and **reason** to analyze and adapt them.

2 Why it Applies to Architecture?



Architecture decisions are rarely about a single correct answer. They are trade-offs made under competing pressures. A human-centered method keeps those decisions balanced, outcome-focused, collaborative, innovative, and grounded in both human and business needs.

The Three Pillars of Design Thinking



3 Internal Mechanism — The Three Pillars in Practice

Pillar	Role in Architecture	Example (Alpha Games)
 Empathy	Understand who uses the data and what outcomes they need	Interview game designers, analysts, and ML engineers at Alpha.
 Creativity	Generate multiple solution paths without premature commitment	Consider streaming, batch, and hybrid ingestion patterns.
 Reason	Analyze trade-offs against ranked business criteria	Compare cost, scalability, and time-to-market for each pattern.

Result

- ✓ Balanced architecture decisions
- ✓ Aligned with business outcomes
- ✓ Multiple solution options evaluated
- ✓ Grounded in real user needs
- ✓ A practical and implementable future-state design

Applying design thinking to build AI-ready, governed data architectures on the Databricks Intelligence Platform.

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The O / \$ / X Classification

Every gap in the current architecture falls into one of three categories.

O

Opportunity

A chance to do something the current architecture cannot support at all. New capability, new use case.

Example: Real-time streaming analytics

\$

Cost / Optimization

The current approach works but is too expensive, too slow, or too fragile. Reduce spend or improve efficiency.

Example: Consolidate 7 separate tools

X

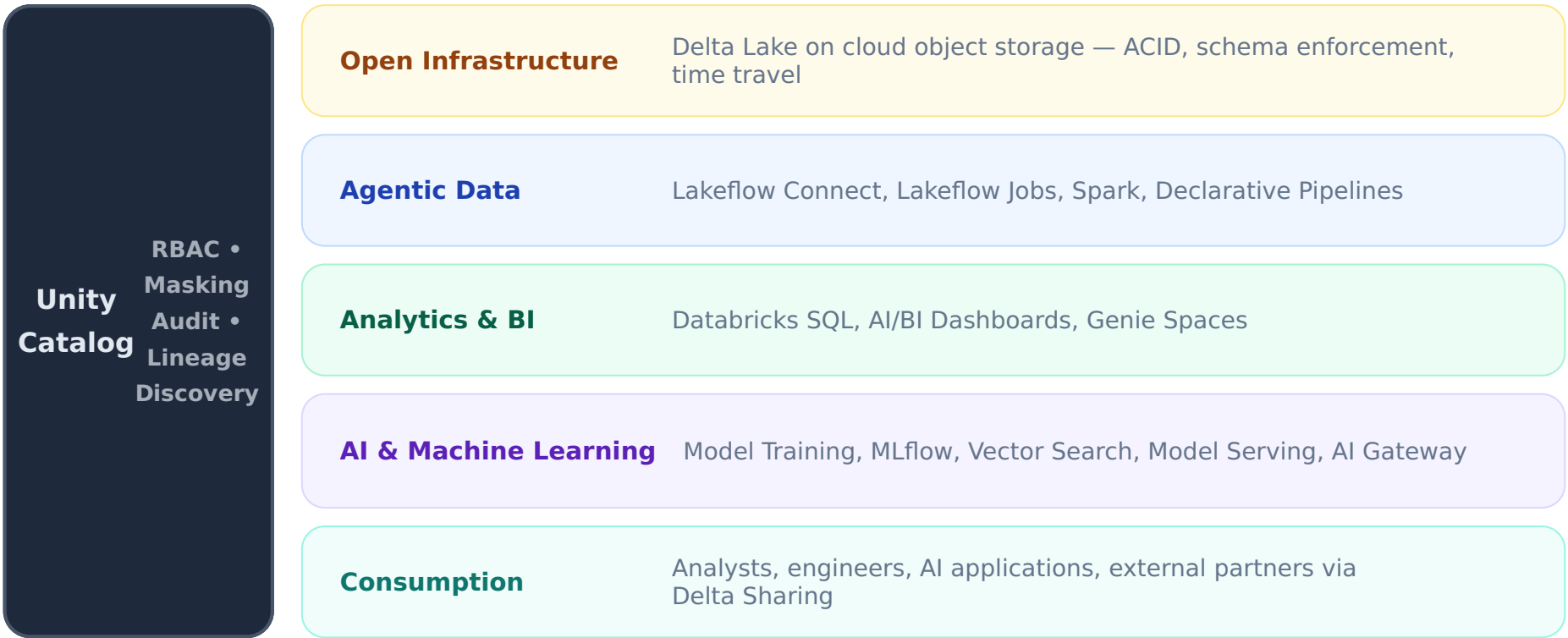
Critical Risk

Something that will fail, cause compliance issues, or block the business. Must be addressed immediately.

Example: No access control on S3 data

The Databricks Intelligence Platform

Five layers governed by Unity Catalog — one platform for data, analytics, and AI.



Four Ingestion Patterns

Each solves a different data-movement problem while keeping everything on one governed platform.

PATTERN 1

Built-in Ingestion

Tool: Lakeflow Connect

Use case: SaaS apps, operational databases

Mechanism: Managed CDC connectors with no infrastructure to maintain.

PATTERN 2

Batch & ETL

Tool: Spark + Auto Loader + Lakeflow Jobs

Use case: File-based loads, scheduled transforms

Mechanism: Incremental file discovery with orchestrated retries.

PATTERN 3

Streaming & CDC

Tool: Structured Streaming + Auto Loader

Use case: Real-time events, database CDC

Mechanism: Exactly-once micro-batch with stateful processing.

PATTERN 4

Lakehouse Federation

Tool: Foreign catalogs in Unity Catalog

Use case: Query external systems without copying data

Mechanism: Push-down queries with governed read access.

Five Consumption Patterns

What you do with data once it is in the lakehouse.

1

Business Intelligence
AI/BI Dashboards + Genie

Replace Tableau with native dashboards on live Delta data with NL querying.

2

Traditional ML
Model Training + MLflow

Train churn-prediction models on unified player data with full experiment tracking.

3

Generative AI Agents
Vector Search + AI Gateway

Build an in-game AI assistant using RAG over documentation with governed endpoints.

4

Data Applications
Databricks Apps

Deploy custom interactive data apps for game designers directly on the platform.

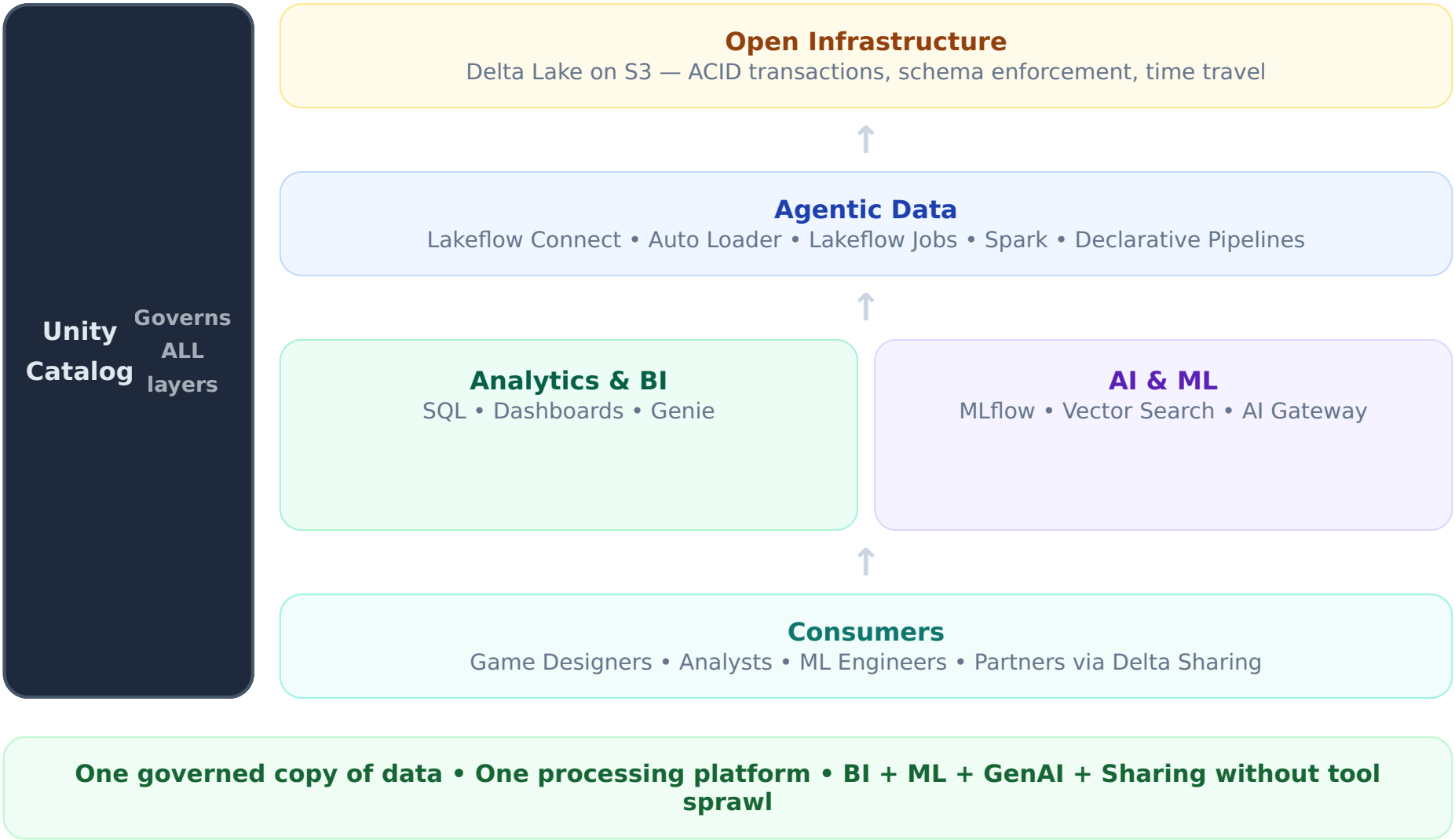
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Data Sharing
Delta Sharing

Share anonymized player data with external research partners — zero copy, governed.

Future-State Architecture

One governed platform replaces seven disconnected tools.



Phased Implementation Roadmap

Prioritized by business value and implementation effort. Delivered in three phases.

PHASE 1
Foundation
Months 1-3

- ✓ Unity Catalog deployment
- ✓ Lakeflow Connect for MySQL
- ✓ Delta Lake migration from S3
- ✓ Access control & audit logging

PHASE 2
Consolidation
Months 4-6

- ✓ Retire Redshift & EMR
- ✓ AI/BI Dashboards replace Tableau
- ✓ Streaming pipeline for game events
- ✓ ML training on unified data

PHASE 3
Innovation
Months 7-12

- ✓ GenAI game assistant (RAG)
- ✓ Real-time player analytics
- ✓ Delta Sharing with partners
- ✓ Databricks Apps for designers

APPLY THE FRAMEWORK

Six Industry Use Cases

Each use case applies the full 6-module workshop to a different industry — same methodology, unique domain challenges.

HEALTHCARE

MedCore Health

12 facilities • Epic EHR • HIPAA compliance

Key gaps: Silent ETL failures, SSIS single point of failure, manual HIPAA audits, no FHIR integration

FINANCIAL SERVICES

Pinnacle Bank

4M customers • AS/400 mainframe • Oracle DW

Key gaps: Mainframe FTP failures, 30% fraud miss rate, manual Basel III reporting

RETAIL

UrbanCart

8M customers • 200K SKUs • Shopify + Snowflake

Key gaps: Airflow DAG failures at peak, no monitoring on Python scripts, static recommendations

MANUFACTURING

TitanMfg

6 plants • 400+ CNC machines • 10K IoT sensors

Key gaps: PI Historian at capacity, manual OEE spreadsheets, no predictive maintenance

TELECOM

NexaComm

Network provider • 5G rollout • Churn analytics

Key gaps: Network optimization, real-time anomaly detection, subscriber churn prediction

ENERGY

GreenGrid Energy

Renewable utility • Smart grid • ESG reporting

Key gaps: Demand forecasting, renewable intermittency, regulatory ESG compliance

Each use case produces: 9 tables • 6-module SQL walkthrough • E2E validation scorecard • Domain-specific gap analysis

EVERYTHING YOU NEED

Workshop Kit & Companion Materials

A complete, portable set of notebooks, exercises, quizzes, and domain-specific use cases.

CORE

Main Workshop Notebook

58 cells • 6 modules • SQL-backed
9 Unity Catalog tables created
E2E validation scorecard
Cross-account portable
(Serverless, Shared, Assigned)

HANDS-ON

Design Exercises

6 structured design exercises
Architecture diagramming challenges
Gap analysis templates
Group discussion prompts

ASSESSMENT

Quizzes & PDFs

3 quizzes × 8 MCQs each (24 total)
SQL answer-check cells
Printable PDF handouts
Stored in Unity Catalog Volume

DOMAIN-SPECIFIC USE CASE NOTEBOOKS

Healthcare

MedCore Health

Finance

Pinnacle Bank

Retail

UrbanCart

Manufacturing

TitanMfg

Telecom

NexaComm

Energy

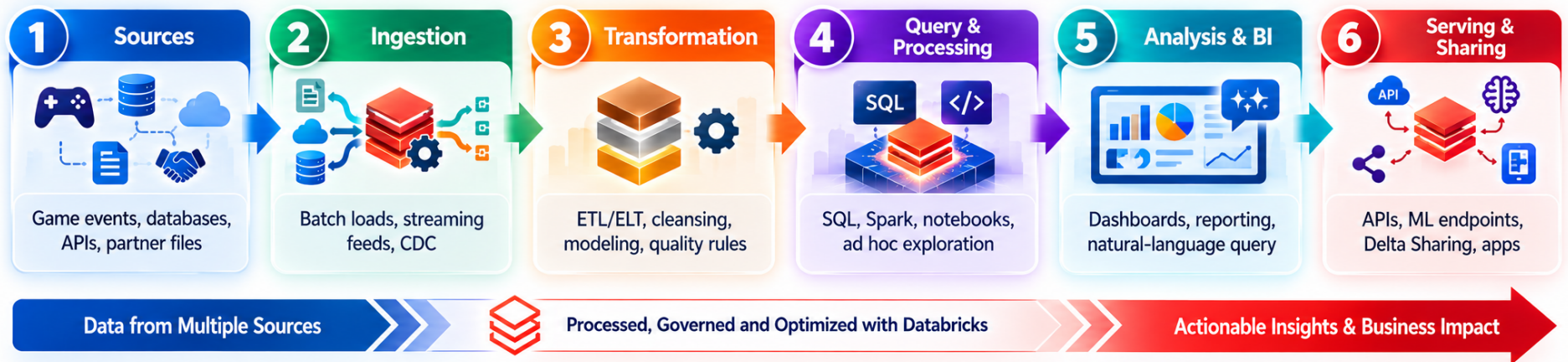
GreenGrid

Each notebook: config cell + Modules 1-6 + 9 tables + E2E validation scorecard • Same portable design



The Data Lifecycle Flow

A consistent end-to-end flow used to document current-state and design future-state architecture



The universal data lifecycle — six stages from source to serving that every architecture must address.

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Your Architecture Starts Here

You now have the framework, the patterns, and the roadmap. The next step is to apply them to your organization using the companion notebooks.

1

Main Notebook

SQL-backed walkthrough
with validation

2

Design Exercises

Hands-on architecture
challenges

3

Knowledge Quizzes

Test your understanding
across all modules

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